

Insights

Analysis Objective
Examine whether depression (PHQ-9), anxiety (GAD-7), and perceived stress (PSS-10) are associated with academic performance (CGPA), and determine whether these mental-health factors remain important after accounting for academic, lifestyle and other student-level factors.

Analytical Approach
EDA → Correlation → VIF → Individual Regression Models → Combined Mental-Health Model → Full Regression Model → Model Comparison

Key Numbers	
KPI	Value
Students analysed	475
GAD-7 ↔ CGPA	-0.334
PHQ-9 ↔ CGPA	-0.307
PSS-10 ↔ CGPA	-0.244
Mental-health model R ²	11.78%
Full model R ²	41.46%
Model improvement	+29.67 pp

Hypotheses

Hypotheses	Conclusion	Inference
H1: Higher depression scores (PHQ-9) are associated with lower CGPA.	Higher PHQ-9 scores are significantly associated with lower CGPA when depression is examined individually. Each 1-point increase in PHQ-9 is associated with approximately a 0.024-point lower CGPA in the individual regression model.	Supported
H2: Higher anxiety scores (GAD-7) are associated with lower CGPA.	Anxiety shows the strongest individual negative association with CGPA among the three mental-health measures. Each 1-point increase in GAD-7 is associated with approximately a 0.033-point lower CGPA in the individual regression model.	Supported
H3: Higher perceived stress (PSS-10) is associated with lower CGPA.	Higher perceived stress is significantly associated with lower CGPA when examined individually. Each 1-point increase in PSS-10 is associated with approximately a 0.016-point lower CGPA .	Supported
H4: Depression, anxiety and perceived stress collectively have a statistically significant association with CGPA.	The three mental-health measures collectively show a statistically significant association with CGPA, explaining 11.78% of observed CGPA variation . However, when all three are considered simultaneously, only GAD-7 remains statistically significant (does not mean depression and stress have no relationship with academic performance).	Supported
H5: Mental-health factors alone account for the major observed variation in academic performance, with other academic and lifestyle factors adding limited explanatory value.	Mental health is not the only set of factors associated with academic performance. Adding academic and lifestyle variables increased the model's explained variation in CGPA from 11.78% to 41.46% , with the improvement statistically significant.	Not Supported

Significant Findings

Finding	Interpretation
Finding 1: Anxiety shows the strongest relationship with CGPA Variable ↔ Correlation with CGPA as GAD-7 ↔ -0.334 PHQ-9 ↔ -0.307 PSS-10 ↔ -0.244	GAD-7 has the strongest negative correlation with CGPA. It also has the highest individual-model R ² : 11.18%. And it remains significant in both the combined mental-health model and the full model.
Finding 2: Depression, anxiety and stress are highly interconnected Correlation of Pairs as PHQ-9 ↔ GAD-7 = 0.872 PHQ-9 ↔ PSS-10 = 0.849 GAD-7 ↔ PSS-10 = 0.807	Depression, anxiety and perceived stress capture overlapping aspects of mental-health burden in this sample, making it difficult to isolate their individual effects in a combined regression.
Finding 3: (Academic Behaviour) Study hours: β = +0.0161; p < 0.001	Higher weekly study hours are positively associated with CGPA after accounting for mental-health and other included factors.
Findings 4: (Academic Behaviour) Attendance: β = +0.00877; p < 0.001	Higher class attendance is positively associated with CGPA and is one of the strongest statistically significant predictors in the full model.
Finding 5: (Lifestyle Behaviour) Daily sleep: β = +0.0649; p < 0.001	Students reporting more daily sleep tend to have higher CGPA after controlling for the other variables in the model. A one-hour increase in reported daily sleep is associated with approximately a 0.065-point higher CGPA. (An association rather than proof)
Finding 6: Part-time Employment Finding: β = -0.1000; p = 0.00052	Students with a part-time job have an estimated 0.10-point lower CGPA than students without a part-time job, holding the other variables constant.

Non-significant Findings

In Model 2		
Variable	p-value	Inference
PHQ-9	0.15	Not significant
PSS-10	0.127	Not significant
Screen time	0.316	Not significant
Physical activity	0.783	Not significant
Financial stress	0.613	Not significant
Interpretation: After accounting for the other variables in the full model, these variables did not show statistically significant independent associations with CGPA at the 5% significance level. The analysis did not find sufficient statistical evidence of an independent association with CGPA.		

Model Comparison

	Mental Health Model	Full Model
Predictors	3	10
R ²	11.78%	41.46%
Adjusted R ²	11.22%	40.19%
N	475	475
ΔR ²	—	+29.67 pp
Model improvement	—	p < 0.001
Interpretation: Academic and lifestyle variables add substantial explanatory value beyond mental-health measures alone.		

Overall Conclusion

Mental health is significantly associated with academic performance, but it is not the only factor associated with CGPA. Depression, anxiety and perceived stress each show significant negative associations with CGPA when examined individually. Among the three, anxiety shows the strongest and most consistent independent association. When academic and lifestyle factors are added, the model's explained variation in CGPA increases from 11.78% to 41.46%. Anxiety remains negatively associated with CGPA, while study hours, attendance and sleep show positive associations and part-time employment shows a negative association. These results suggest that academic performance is associated with a combination of mental-health, academic behaviour and lifestyle factors, rather than mental health alone. The analysis identifies statistical associations, not causal relationships.